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Supplemental Information

HORMA Domain Proteins and a Trip13-like ATPase

Regulate Bacterial cGAS-like Enzymes

to Mediate Bacteriophage Immunity

Qiaozhen Ye, Rebecca K. Lau, Ian T. Mathews, Erica A. Birkholz, Jeramie D. Watrous, Camillia S. Azimi, Joe Pogliano, Mohit Jain, and Kevin D. Corbett

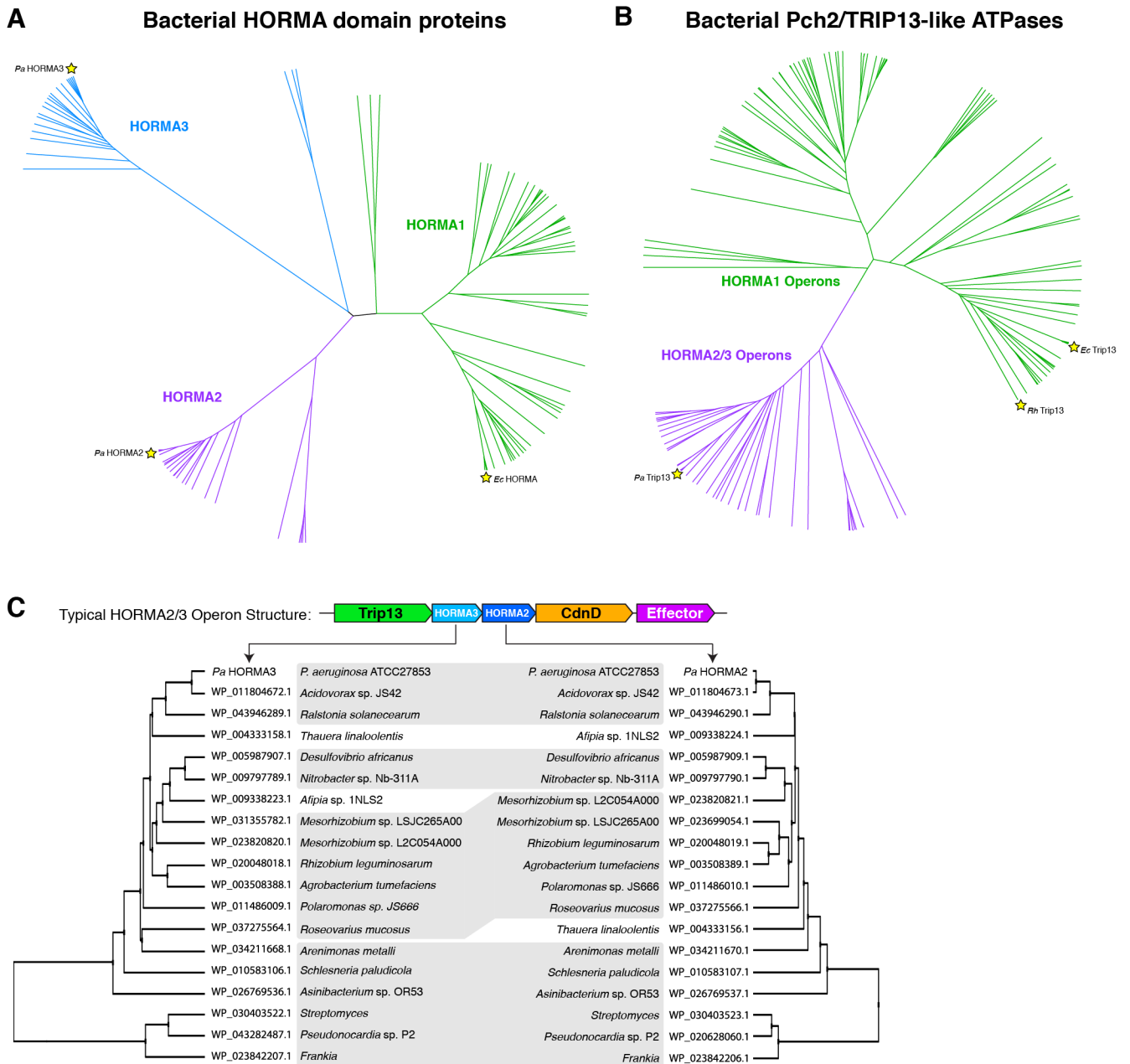


Figure S1 (related to Figure 1). Sequence analysis of bacterial HORMA proteins and Pch2/TRIP13-like ATPases

(A) Phylogenetic tree of bacterial HORMA domain proteins, with HORMA1 proteins colored green, HORMA2 proteins purple, and HORMA3 proteins blue. Stars indicate proteins used in this study: *Ec* HORMA, *Pa* HORMA2, and *Pa* HORMA3. Prior studies (Burroughs et al., 2015; Tromer et al., 2019) identified two families of bacterial HORMA proteins, but our analysis indicates the existence of three families. HORMA1 proteins are encoded in operons with one HORMA domain protein, while HORMA2 and HORMA3 proteins are encoded together (see panel C). **(B)** Phylogenetic tree of bacterial Pch2/TRIP13-like ATPases, with proteins from HORMA1 operons colored green, and proteins from HORMA2/3 operons colored purple. Yellow stars indicate proteins used in this study: *Ec* Trip13, *Pa* Trip13, and *Rhizobiales* (*Rh*) Trip13. The divergence of Trip13 proteins from HORMA1 and HORMA2/3 operons agrees with a prior report (Tromer et al., 2019), and suggests that HORMA and Trip13 proteins have co-evolved within each operon family. **(C)** Phylogenetic trees for 19 pairs of HORMA2 and HORMA3 proteins annotated by Burroughs et al. (Burroughs et al., 2015), showing co-evolution of the two protein families within the operon. See **Table S1** and **Table S2** for all HORMA and Trip13 sequences used to generate panels A, B and C.

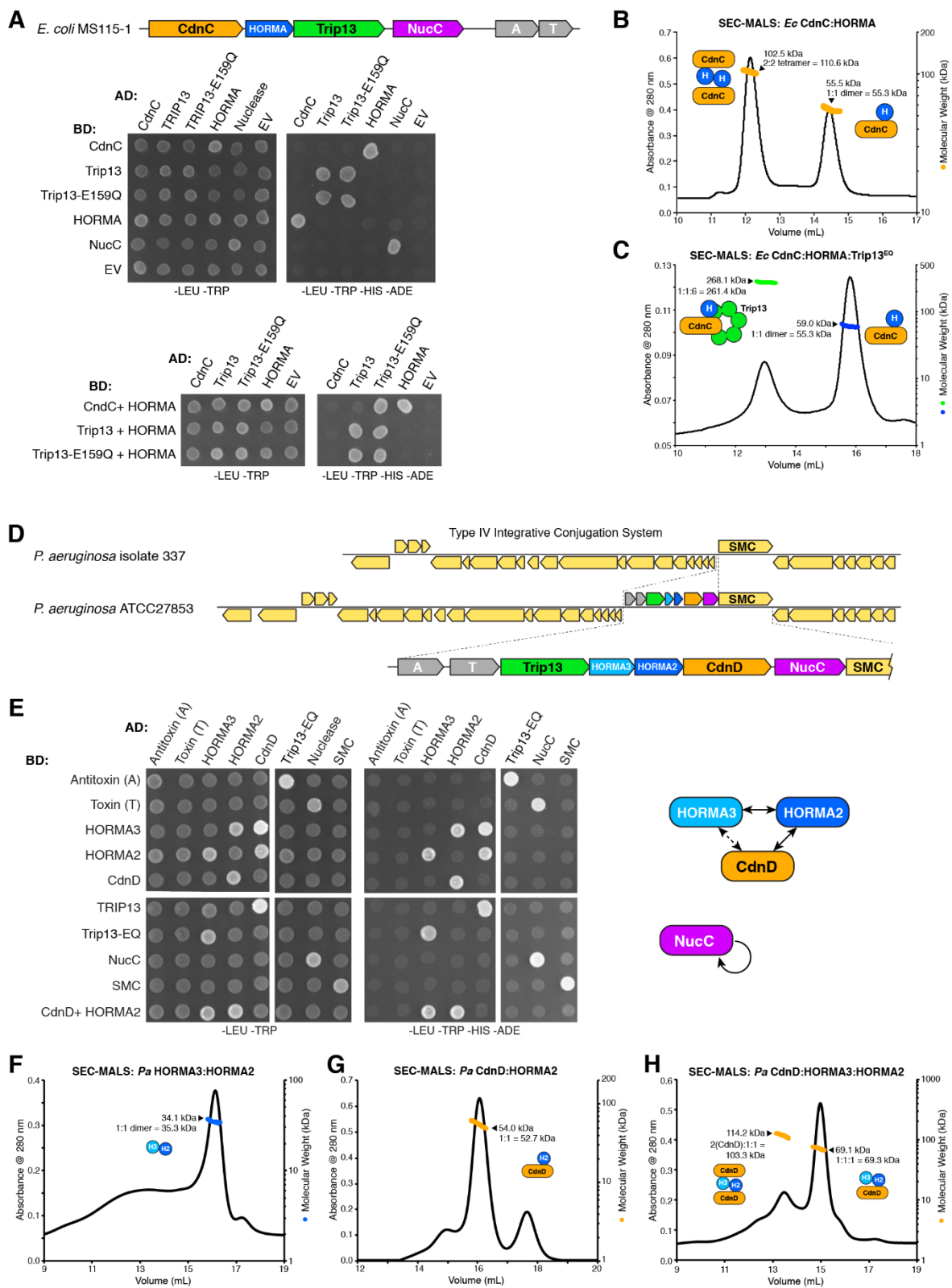


Figure S2 (Related to Figure 1). Protein-protein interactions in HORMA/Trip13-associated CBASS systems.

(A) Yeast two-hybrid and three-hybrid analysis of protein-protein interactions in the *E. coli* MS115-1 CD-NTase containing operon. “-LEU -TRP” refers to non-selective growth media, and “-LEU -TRP -HIS -ADE” refers to selective growth media. This operon encodes a RES-Xre type toxin-antitoxin gene pair, labeled A/T, which was not tested for protein-protein interactions. “EV” refers to empty vector controls. **(B)** SEC-MALS of the *Ec* CdnC:HORMA complex, showing that it forms 1:1 and 2:2 complexes in solution. **(C)** SEC-MALS of *Ec* CdnC:HORMA:Trip13^{EQ}, showing that it forms a 1:1:6 complex in solution when pre-incubated with ATP. **(D)** Schematics comparing the genomic context of the *P. aeruginosa* ATCC27853 CBASS system to a homologous locus in *P. aeruginosa* isolate 337 (NCBI reference sequence NZ_FRQLO1000018.1). The CBASS system in *P. aeruginosa* ATCC27853 has been incorporated into a putative Type IV integrative conjugation system (yellow) that encodes a

putative SMC-like ATPase. While the CBASS system is encoded on the same strand as the putative SMC protein, this protein is not expected to function with the CBASS system. **(E)** Yeast two-hybrid and three-hybrid analysis of protein-protein interactions in the *P. aeruginosa* ATCC27853 CBASS operon. In addition to the CBASS proteins (CdnD, HORMA3, HORMA2, Trip13, and NucC), the putative toxin-antitoxin gene pair (PIN domain protein (T) and helix-turn-helix protein (A)) and SMC-like ATPase were tested. *Right:* Summary of protein-protein interactions in the *P. aeruginosa* ATCC27853 operon. The dashed line between HORMA3 and CdnD indicates that this interaction was detected with only one tag combination; nonetheless the two proteins form a complex when co-expressed in *E. coli* (not shown). **(F)** SEC-MALS analysis of *Pa* HORMA3:HORMA2, showing that the proteins form a 1:1 complex in solution. **(G)** SEC-MALS analysis of *Pa* CdnD:HORMA2, showing that the proteins form a 1:1 complex in solution. Size-exclusion chromatography of purified *Pa* CdnD:HORMA3 complex indicates a likely 1:1 stoichiometry, though aggregation of the complex precluded analysis by SEC-MALS (not shown). **(H)** SEC-MALS analysis of *Pa* CdnD:HORMA3:HORMA2, showing that the proteins form a mix of 1:1:1 and 2:1:1 stoichiometries in solution.

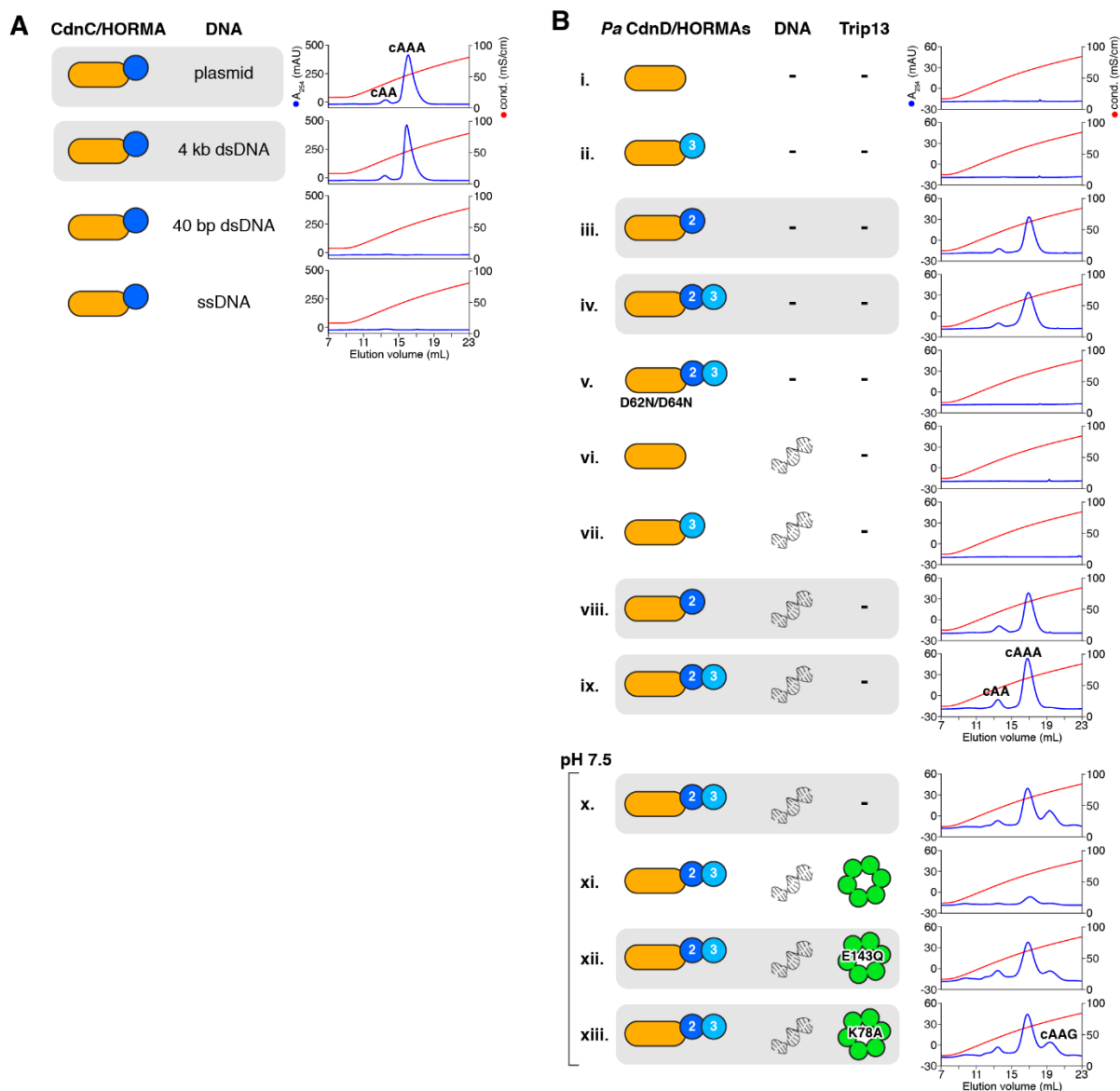


Figure S3 (Related to Figure 1). Second messenger synthesis by *Ec* CdnC and *Pa* CdnD

(A) Second messenger synthesis assays as in **Figure 1C**, with different DNAs: ~4 kb plasmid DNA (top), 4 kb linear dsDNA, 40 bp linear dsDNA, and single-stranded DNA (sheared salmon-sperm DNA). **(B)** Second messenger synthesis assays with *Pa* CdnD, HORMA3 (light blue), HORMA2 (dark blue), plasmid DNA, and Trip13. The peaks labeled “cAA” (sample ix), “cAAA” (sample ix), and “cAAG” (sample xiii) were verified by LC MS/MS. Samples x-xiii were assayed at pH 7.5 rather than 8.5 (as in samples i-ix) to accommodate *Pa* Trip13, which was insoluble at pH 8.5. *Pa* CdnD synthesizes cAAG as a minor product at pH 7.5.

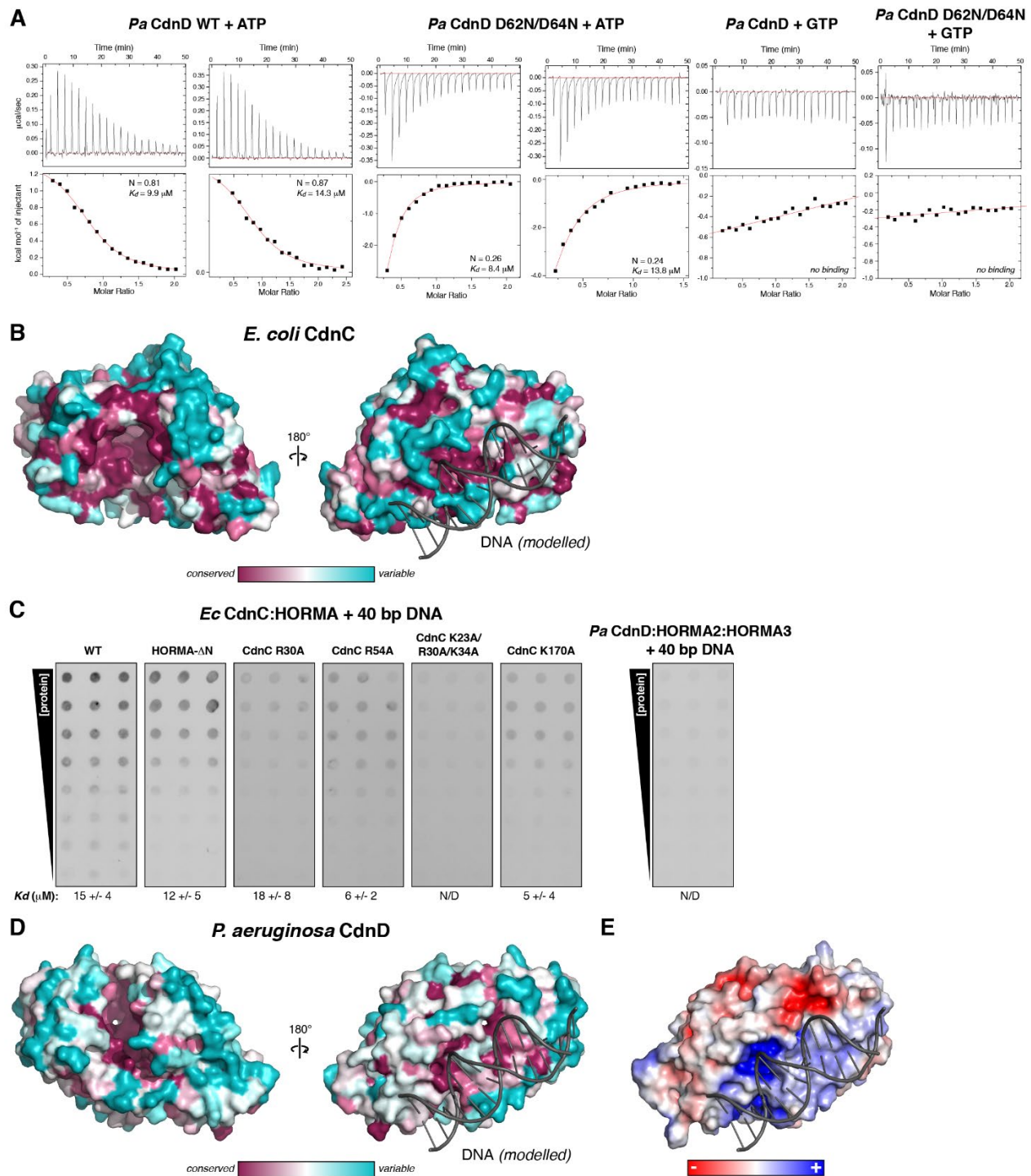


Figure S4 (Related to Figure 2) - Nucleotide and DNA binding of *Ec* CdnC and *Pa* CdnD

(A) Isothermal titration calorimetry (ITC) analysis of *Pa* CdnD binding either ATP or GTP. Both wild-type and D62N/D64N mutants bind ATP with a K_d of ~ 10 μ M, but neither detectably binds GTP. A similar analysis of *Ec* CdnC was not possible due to its stably-bound ATP. (B) Two views of *Ec* CdnC (left panel equivalent to **Figure 2A**), with surface colored by conservation. Bound DNA (gray) is modelled from a superposition with *M. musculus* cGAS bound to DNA (PDB 4K9B) (Gao et al., 2013). (C) DNA binding dot-blot assays for *Ec* CdnC:HORMA and *Pa* CdnD:HORMA2:HORMA3 complexes. Panels show triplicate binding assays with a 5'-fluorescein-labeled 40 bp dsDNA, with protein concentration starting at 50 μ M and dropping 2-fold each row (final row, 0.39 μ M). K_d values were calculated with PRISM; N/D, not determined. (D) Two views of *Pa* CdnD equivalent (left panel equivalent to **Figure 2B**), with surface colored by conservation. (E) View of *Pa* CdnD colored by surface charge.

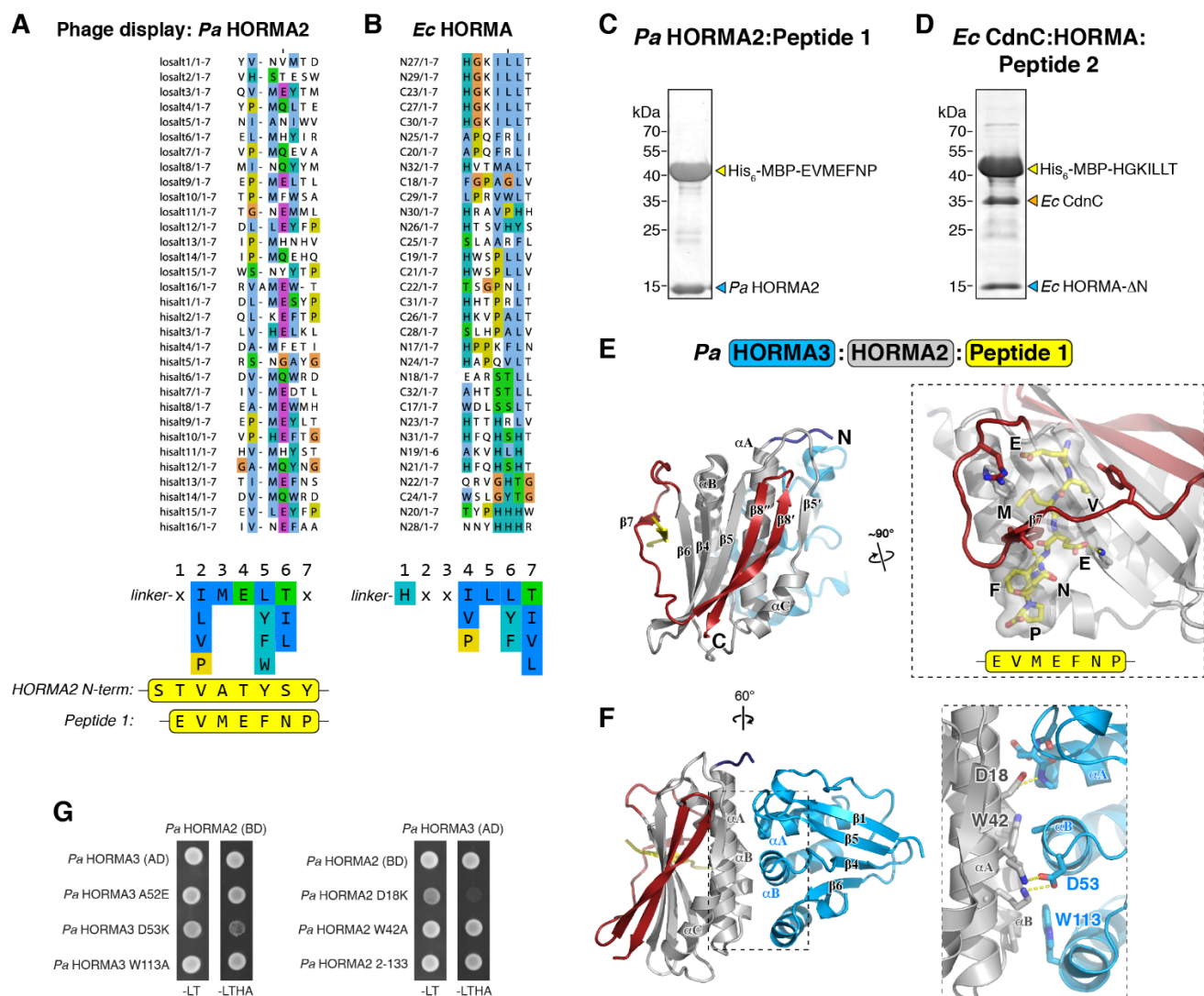


Figure S5 (Related to Figure 3). HORMA-closure motif and HORMA-HORMA interactions

(A) Aligned sequences from round 4 of phage display with *Pa* HORMA2 as bait. A consensus sequence is shown at bottom, aligned with the sequence of the *Pa* HORMA2 N-terminus, which binds *Pa* HORMA2 as a closure motif in our crystal structure (Figure 3B), and Peptide 1, designed from the consensus motif. (B) Aligned sequences from round 4 of phage display with *Ec* HORMA as bait. A consensus sequence is shown at bottom. (C) Ni-NTA purification of coexpressed His₆-MBP-Peptide 1 (sequence: EVMEFNP) with untagged *Pa* HORMA2. (D) Ni-NTA purification of coexpressed His₆-MBP-Peptide 2 (sequence: HGKILLT) with untagged *Ec* HORMA-ΔN and *Ec* CdnC. (E) Structure of the *Pa* HORMA3:HORMA2:Peptide 1 complex, with closeup view of HORMA2 binding Peptide 1 (yellow). (F) Rotated view of the *Pa* HORMA3:HORMA2:Peptide 1 complex, with closeup showing the HORMA1-HORMA2 interface. (G) Yeast two-hybrid analysis of the *Pa* HORMA1-HORMA2 interaction. BD: fusion to the Gal4 DNA-binding domain; AD: fusion to the Gal4 activation domain. -LT: media lacking leucine and tryptophan (non-selective); -LTHA: media lacking leucine, tryptophan, histidine, and adenine (stringent selection).

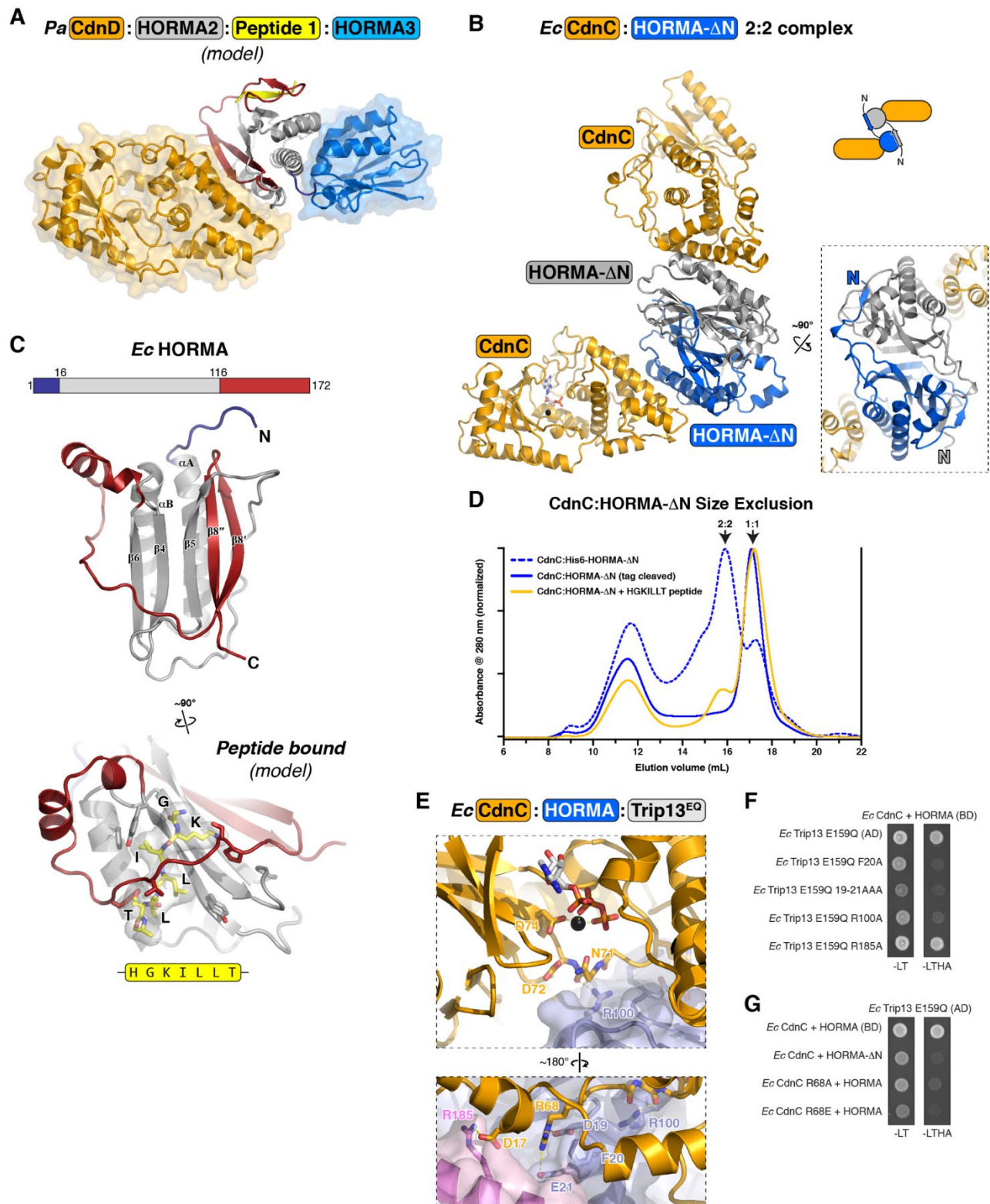


Figure S6 (Related to Figures 4 and 5). Structure of CD-NTase:HORMA complexes

(A) Model of a *Pa* CdnD:HORMA2:Peptide 1:HORMA3 complex, based on separate structures of CdnD:HORMA2:Peptide 1 (Figure 4A) and HORMA3:HORMA2:Peptide 1 (Figure 3E). HORMA3 and CdnD occupy different surfaces of HORMA2, suggesting they can bind simultaneously. (B) Overall view of the asymmetric unit in crystals of *Ec* CdnC:HORMA-ΔN. The N-terminal His6-tag of each HORMA-ΔN protomer mimics a closure motif in its non-crystallographic symmetry mate (see inset; two HORMA-ΔN protomers colored blue and gray), forming a symmetric 2:2 complex. (C) Top: Overall structure of *Ec* HORMA in the *Ec* CdnC:HORMA-ΔN complex. Coloring is equivalent to Figure 3A-D, with the mobile N-terminus blue and safety belt red. The protein adopts a closed, but empty, conformation. Bottom: Model of *Ec* HORMA bound to a consensus closure motif peptide (HGKILLT), based on a superposition of *Ec* HORMA-ΔN with the *Pa* HORMA2:Peptide 1 complex. (D) Superdex 200

size exclusion chromatography elution profiles for *Ec* CdnC:HORMA- Δ N with N-terminal His₆-tag on HORMA intact (dotted blue line), cleaved (solid blue line), or cleaved in the presence of Peptide 2 (HGKILLT; orange). Prior to tag cleavage, the complex adopts predominantly 2:2 stoichiometry. After cleavage, the complex adopts predominantly 1:1 stoichiometry. **(E)** Closeup views of the *Ec* CdnC-Trip13^{EQ} interface in the *Ec* CdnC:HORMA:Trip13^{EQ} complex, with CdnC orange and Trip13^{EQ} monomers E and F shown as light blue and violet, respectively. *Top* shows closeup view equivalent to that of panel A, and *bottom* shows opposite view. **(F)** Yeast three-hybrid analysis (*Ec* Trip13 in pGADT7, *Ec* CdnC (AD fusion) and HORMA (untagged) in pBridge) showing that mutation of Trip13^{EQ} residues Asp19, Phe20, Glu21, and Arg100 disrupt binding to CdnC. -LT: media lacking leucine and tryptophan (non-selective); -LTHA: media lacking leucine, tryptophan, histidine, and adenine (stringent selection). **(G)** Yeast three-hybrid analysis showing that mutation of *Ec* CdnC residue Arg68, or truncation of the HORMA N-terminus, disrupt binding to Trip13^{EQ}.

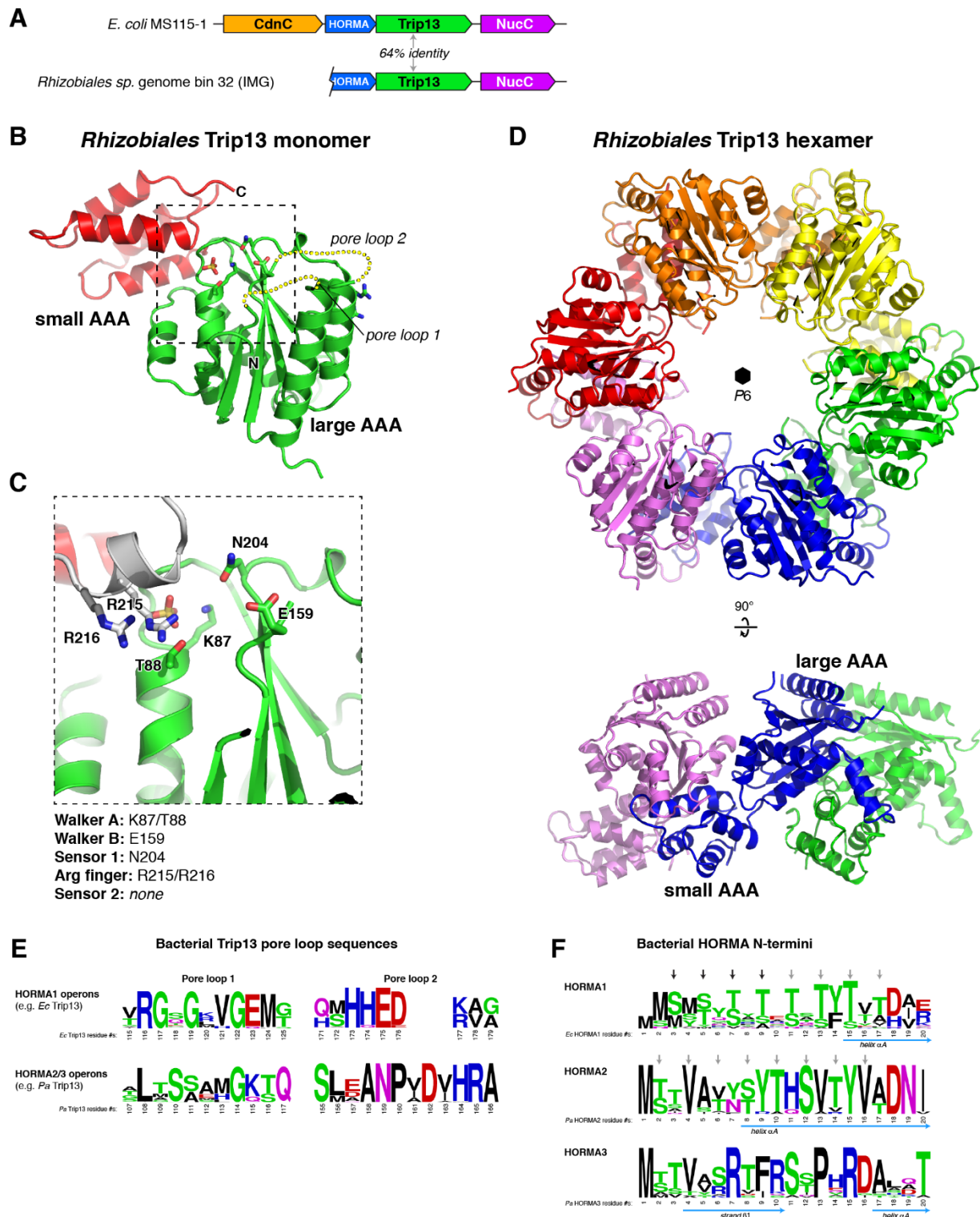


Figure S7 (Related to Figure 5). Structure of *Rhizobiales* Trip13

(A) Schematic of the *Rhizobiales* CBASS system, including HORMA (IMG #2619783695), Trip13 (IMG #2619783694, sequence in **Table S4**), and NucC (IMG #2619783693), compared to the *E. coli* MS115-1 operon. The sequenced contig at IMG (JGI Integrated Microbial Genomes database) ends at the putative HORMA protein, precluding identification of the inferred CD-NTase. **(B)** Overall structure of the *Rhizobiales* (*Rh*) Trip13 monomer, with N-terminal “large AAA” domain green and C-terminal “small AAA” domain red. Yellow dotted lines indicate the positions of pore loops 1 and 2, both disordered in this structure. **(C)** Closeup view of the *Rh* Trip13 active site, in the same orientation as in panel B. A SO_4^- ion is bound in the active site, surrounded by canonical Walker ATPase active site residues. Bacterial Trip13 proteins from 1-HORMA operons possess a conserved double Arg finger (R215/R216; from neighboring subunit, colored white); see **Figure S8F** for sequences. **(D)** Top and side views of the six-fold symmetric *Rh* Trip13 hexamer, formed from crystallographic packing interactions (P6 symmetry axis noted by black hexagon). **(E)** Sequence logos showing divergence of pore loop sequences in Trip13 proteins from one-

HORMA and two-HORMA operons. **(D)** Sequence logos showing the N-termini of HORMA proteins from one-HORMA operons (top), and from the two families of HORMA proteins from two-HORMA operons (middle/bottom). The first residue of helix α_A (based on crystal structures of *Ec* HORMA, *Pa* HORMA2, and *Pa* HORMA3) are indicated by blue arrows. Arrows above the sequence of *Ec* HORMA (top) indicate serine/threonine residues directly engaged by *Ec* Trip13 pore loop 2 residue His173: black arrows indicate residues engaged by pore loops in our structure, and gray arrows indicate continuation of the conserved pattern into helix α_A .

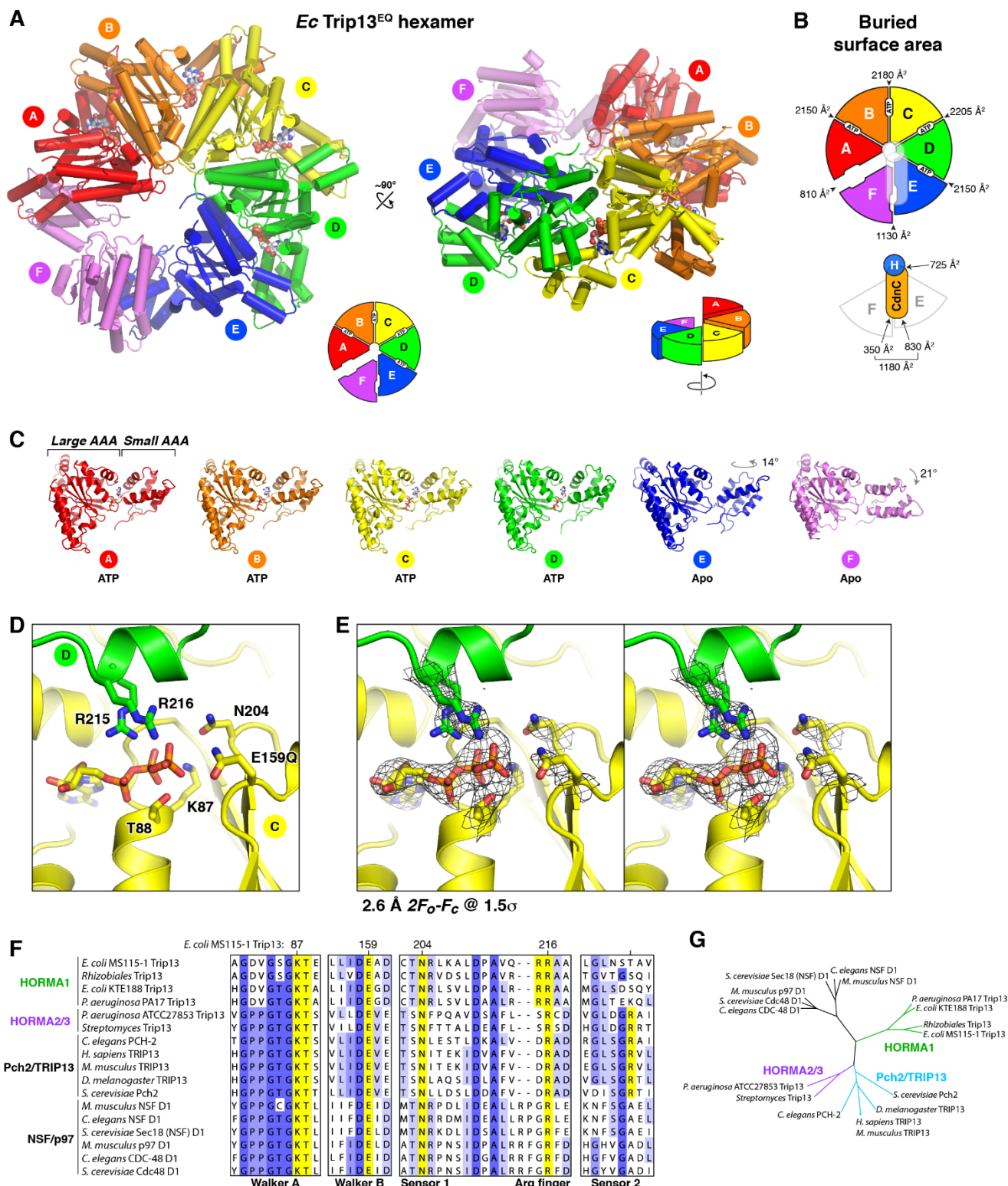
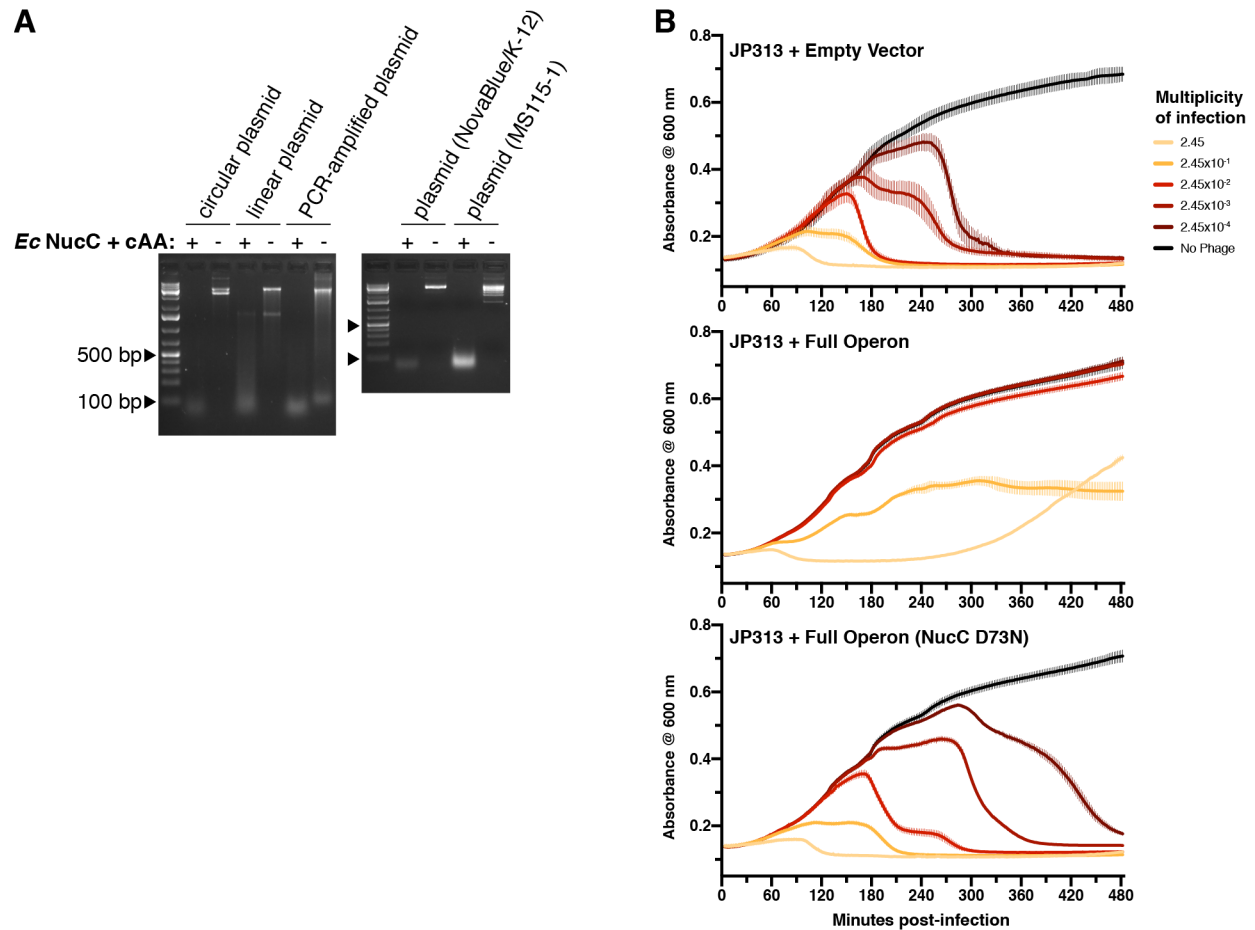


Figure S8 (Related to Figure 5). Structure of the *Ec* CdnC:HORMA:Trip13^{EQ} complex

(A) Top and side views of the *Ec* Trip13^{EQ} hexamer in the structure of CdnC:HORMA:Trip13^{EQ}, with subunits A-F colored red, orange, yellow, green, blue, and violet, respectively. Bound ATP molecules at the A-B, B-C, C-D, and D-E interfaces are shown as spheres. **(B)** Schematics of the *Ec* CdnC:HORMA:Trip13^{EQ} complex, showing buried surface area in each subunit interface: inter-Trip13 interfaces top; Trip13-CdnC and CdnC-HORMA interfaces bottom. **(C)** Views of each individual subunit in the *Ec* Trip13^{EQ} hexamer, aligned on the large AAA domain, showing the conformational similarity of subunits A-D, and rotation of the small AAA domain in subunits E and F. **(D)** Closeup of ATP binding to *Ec* Trip13^{EQ} (subunit C), with conserved ATP-interacting residues from chain C (Walker A, Walker B, Sensor 1) in yellow and from chain D (double Arg finger) in green. **(E)** Stereo view of refined 2F_o-F_c electron density, contoured at 1.5 σ, surrounding ATP and ATP-interacting residues in Trip13^{EQ} chain C. **(F)** Conserved AAA+ sequence motifs in bacterial Trip13, eukaryotic Pch2/TRIP13, and eukaryotic NSF and p97

proteins (D1 ATPase domain). While Trip13 proteins from HORMA2/3 operons (e.g. *P. aeruginosa* ATCC27853 Trip13) share all motifs with eukaryotic Pch2/TRIP13 proteins, Trip13 proteins from HORMA1 operons (e.g. *E. coli* MS115-1 Trip13) possess a double Arg finger (*Ec* Trip13 residues 215 and 216) and lack a sensor 2 arginine. **(G)** Phylogenetic tree showing that bacterial Trip13 proteins from HORMA2/3 operons are more closely related to eukaryotic Pch2/TRIP13 than those from HORMA1 operons.



Methods S1 (Related to STAR Methods). Sequence of *E. coli* MS115-1 CBASS system cloned into plasmid pLAC22 for plaque assays.

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AGCAAGAGCGACCGAATTTTCGCTTTAAATCGTTTCATGATTGAATTGATTTCTTGCAAATTAAGGATAATGGTGTGGATTTCTCGAATTATCCGGAAGCTTTACAGGCATTTT
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CCTATGGAAAATGGTTGGCTAATTGA¹

¹Color-coding: Red: CdnC / Blue: HORMA / Orange: HORMA/Trip13 overlap / Green: Trip13 / Purple: NucC

Table S4 (Related to STAR Methods). Protein Sequences

Protein	NCBI Accession	Sequence
<i>E. coli</i> MS115-1		
¹ CdnC	EFJ98156	MSSGLDRVKTSSDE MSTEHVDHKTARFAEDKVNLPKVKADDFREQAKRLQNKLEGYLSHPDFSL KRMIPSGSLAKGTALRSLNDIDVAVYISGSDAPQDLRGLLDYLADRLRKAFPNFSPDQVKPQTYSVTV SFRGSGLDVDIVPVLYSGLPDWRGHLISQEDGSFLETSLPLHLDFIKARKRAAPKHFQAQVRLAKYWA RLMKQERPNNFRKFSMIELILAKLLDNGVDFSNYPEALQAFFSYLVSTELRERIVFEDNYPASKIGTSLSDL VQIIDPVNPNVNNVARLYTQSNVDAIIDAAMDAGDAIDAAFYAPTQKLTVTYWQKVFSGSSFQG
² HORMA	KDS98063	MSSYSYTAETQTFSVTHARHMAAKVATDLRRMQRFGYGPSDADIEAYEEELVVFLKAGYLGEVSYG FQKNNNWIEPTLRYTAGDLLSGTDDDPGKIRPGKDVSGASFYSFMTYSSKYLNATQSEKDTALKDL PFKRVGAQSPGINGYLENDKTYAGGRSLTRTSVRNFV
Trip13	EFJ98158	MNVKPSLDELFERRINFPDFEPQERLARLVGLDEHKDRLSKILGLLVNPGIQEWAKKYHPDARAAMD TVLRRPPLVVLGADVSGSKTELAETIGDAVARQEDIDITLYPLSLATRGQGRVGEMTQLVSAAFDYTIE AADKLKNTNGKARGAVLLIDEADALAQSRENAQMHEDRAGVNAFIRGIDRIANQKLPAAVLMCT NRLKALDPAVQRRAAEILTSRPNDQRHYLLHSLKTLGLGLNSTAVEELVRLTGPRDPNSPGFTFSDIT QRLIPSIIAAYPYNAVSVHSALQVVKMTPTPAFIDRQ
NucC	EFJ98159	MSDWSLSQLFASLHEDIQLRLGTARKAFQHPGAKGDASEGVWIEMLDTYLPKRYQAANAFVVDLSG NFSQIDVVVFDRQYSPFIKFNEQIIVPAESVYAVFEAKQSASADLVAYAQRKVASVRRHLRHTSLPIP HAGGTYPAKLPIPLGGLTFESDWSPALGMSFDKALNGDLSGRLDMGCVASHGHFYFNNDISKFN FEHGNKPATAFLRLIAQLQFSGTVPMIDIDAYGKWLAN
<i>P. aeruginosa</i> ATCC27853		
Antitoxin (helix- turn-helix protein)	WP_003050245	MTTATAQSKMTLPAAGEVKAAVQGQRALAAYLATQFETQHIQIFDDHKQAHQVELPTSALRLLVDIL AELADGNAVKKVPVHAELTTQEADLLNVSRPHFVKLEDDGVLAFHRTGKHRRVRFADLMQYKEAR ERASEQAMAELAQQSQELGMGYE
Toxin (PIN domain containing protein)	WP_003153636	MRHSPFTAVYDACVLYPAPLRDLFWMWLGLSGRFRARWSQAIHEEWKRNLLINRPDLTRVQVDRDTS LMDRAIPDGLVEGYEALVAGLTLPDPNDRHVLAAAIRCGASVIVTFNERDFPNLLAPYIGIESQHPDE FVDNLLDLAAAVVSAAQQRQAQLKHPPIDVDRYLEILLRQGLVQTTKVLATYRTIL
Trip13	WP_016852809	MTKNPSSDATLPKGIHRSWKLPDKSLGDLWDSIVMDEAIKKQLLSQAIWNFTVRPKVERTVLPLHGV LLVGPPGTGKTSARGLAHRVAESFSSAKFRLLLEVEPHTLTSSAMGKTQRAVADLFSQSIASAAAGPT IVLLDEVETLAADRAKLSLEANPVDVHRATDAVLVQLDMLAERNPHLLFVATSNFPQAVDSAFSLRC DMVMEVPLPGKDACKQILVDCNLGLAKTFPGIGKLSSAHQFDVCAGECVGLDGRAIRKVVANALAA DPQVAIDPNKLSVEHLRSAIRQAKQMRQLQGGKQK
HORMA3	WP_003090160	MTTVVSRTFRSSPHRDALQTWDAIVELLTQKGDTARSELRAVTGVAASLIADQAPKSAPIVATCDG PRTRICYCLFDEDAIDGDDANEVLGFEPKGDWGVSLPCPKEQLGWVQSALKKHSSRIIARDLSQGI TQAQADAGQALSDDLGGFLKS
HORMA2	WP_003090159	MSTVATYSYTHSVTYVTDNILSKLDIILSGLDPEHFADRWESNTRAIKTWLTGDLRKVILEIYNPAT DKLVTRWDIDIVYGWSDGDSFWTDTEQLKYAIKKAGLLPSQAKYKMLMDTKGRPDVEGWSKGSY RSTDGMVKQSLGSTVEHSLAGQAGYWRQR
CdnD	WP_003090158	MLSIDEAFRKFKSRLNERNREQNASQRQNEVRDYLTQKFGIARSFLTGSYARYTKTKPLKDIDIFFVLK DSEKHYHGKAASVVLDDFHSALVEKYGSAVRKQARSINVDGVDHIDAEDNTDYRVVSVDAVPAFD TGDQYEIPDTASGKWIKTDPEIHKDKATAAHQAYANEWKGLVRMVKYWNNNPKHGDLPVKPSFL IEVMALECLYGGWGGSFDRFIQSFFATLADRVHDEWPDAGLGPAINMDAARKQRAQQLLFQA SQDASIAIDHARRGRNIEALRAWRALFGPKFPLS
NucC	WP_003050273	MSQWSLSQLLSSLHEDIQRLSVVRKTFGHPGTGKGDASENVWIDMLDTYLPKRYQAAKAHVVDLSG NFSQIDVVVFDRQYSPFIFTYENETIIPAESVYAVFEAKQTADAGLVAYAQEKVASVRRHLRHTSLPIP AGGTYPAKLPIPLGGLTFESEWSPALGPSMDKALNANLTEGRDIDGCVAAHGHFFYDQASGAYS YTNENKPATAFLFKLIAQLQFSGTVPMIDVEAYGQWLTK
SMC	WP_003050276	MASIRSEYHRFLAHLAQRHVHDDVRRLAHLVLDHLQPLAEVGAARRGRSTRAPLAIHAHQMPVA YDGDARGPENGPAALGRHLQLEVGPFGRGFMREQTFDLSHDITLVYGANGTGKSSFCFALEVAMLSIS EAQAKRVDQRTYCINNARLRHVAPVLSSTAAGEAQAVQPDEAEYRFCIEKNRLDDFARIAARTPSD QRQLIATLFGVDQFSEFVRGFNPSLDQDLMLAGVQAAQLAQRRLRLASSEQTIAAYPQKIAAVEGLE QALAQRMSPGATYQACVDWLLGTPQQQGRLPYVQAQLDANPPAIHEVTQARLQALLAEAYRVQG LWQASSAQLAARAGEVSYAKLYEAVQALADGATACPACGTGLAAVAQDPFARARMGLEQLAQLAV LQQQEAGHRTQLSEAVRALWDEMRRVVAAGVACPAESQAAGLPLLPPTSAGNWLGGWVNGD QRAWQALLRIAQIIIEGFDAQARDVHAQRGAMAQERDRLQHQLEIERLRTMRTTADQELAAARQT VAQFDDANRGLIAATDEMPVVVHHQRVKAAYDGLPEIQAYLTALPGVLLQGLGDQARHLYNAF

		NRADPPGDLLHALWLPVAENGKIEVEFAGEPGVRYDALIVFSEGHKCLGLAILLAKNLAQGCPVVIFD DVVNAIDDDHRDGIWRTFFEDGLLHGKQVILTSHAEEFLHRIQQELGVRRAAAIKRYKFLPHQGEHEL RVDSDDPPAKNYVLLAQQAALAADEKREALRQARPALESLDRLWTWLGRRADGRIDIKLSGPRAPWE LNNKCTKLRSAYERIAAQHAGAPDAVGALVRLNVSQTSIEWGYLNSGVHDAQRDHEFDRATVRTV VEAVTALDAALDTLQNR
<i>Rhizobiales</i> Trip13	2619783694 (see footnote 2)	MSMTPDLAGLFEGASTYPDVDARERLNNLVGLDTHKSRLSKMLAVLVNPDGLSAWAKKHHHPAAEA LVKNVIRRPPLIVLAGDVGSGKTELAETIGDDVARRESIRITLLPLSLSARGQGRVGEMTQLISAAFEHT VSEARKLKSSGKARGAVILLVDEADALAQSREAAQMHHEDRAGVNAFIRGIDRLGNGALPAAVIM CTNRVDSLDPVRRRAAEIITFDRPNDAQRRAVITTTLQGTGVTGSQIEGLVAATGPAPRDYGTFTS DLTQRLIPSIVLDAYPDTSINPARALAIQAAMAPTAPFQDKSR

¹Sequence underlined in red is annotated, but not part of the protein

²Accession number is from the Joint Genome Institute (JGI) Integrated Microbial Genomes & Microbiomes database
(<https://img.jgi.doe.gov>)